PHYSICS

CAPACITANCE_MEGA_XII



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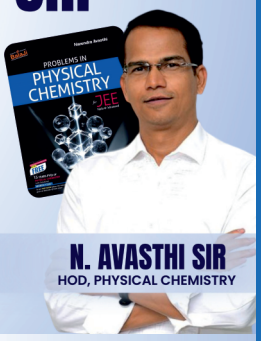

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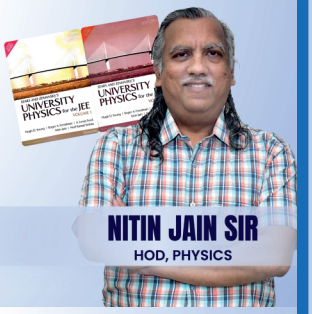


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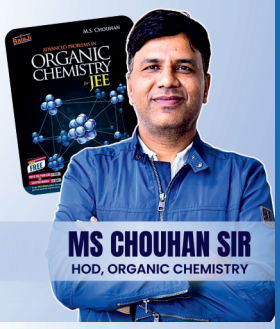


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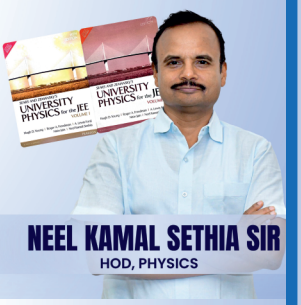


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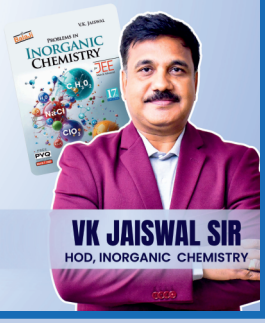


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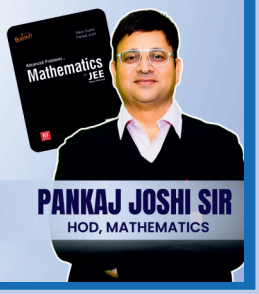


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PHYSICS

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MEGA-XII

CAPACITANCE

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KEY CONCEPTS

1. CAPACITANCE OF AN ISOLATED SPHERICAL CONDUCTOR :

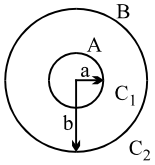
$C = 4\pi \in_0 \in_r R$ in a medium $\qquad C = 4\pi \in_0 R$ in air

* This sphere is at infinite distance from all the conductors .
* The capacitance $C = 4\pi \in_0 R$ exists between the surface of the sphere & earth .

2. SPHERICAL CAPACITOR :

It consists of two concentric spherical shells as shown in figure. Here capacitance of region between the two shells is C_1 and that outside the shell is C_2. We have

$$C_1 = \frac{4\pi \in_0 ab}{b-a} \qquad \text{and} \quad C_2 = 4\pi \in_0 b$$



Depending on connection, it may have different combinations of C_1 and C_2.

3. PARALLEL PLATE CAPACITOR :

(i) UNIFORM DI-ELECTRIC MEDIUM :

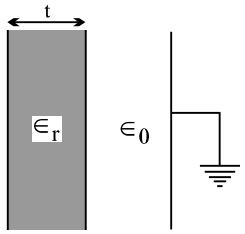
If two parallel plates each of area A & separated by a distance d are charged with equal & opposite charge Q, then the system is called a parallel plate capacitor & its capacitance is given by,

$$C = \frac{\in_0 \in_r A}{d} \text{ in a medium} \quad ; \quad C = \frac{\in_0 A}{d} \text{ with air as medium}$$

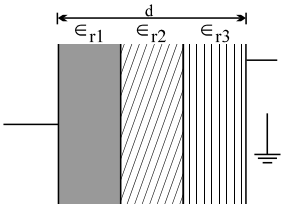
This result is only valid when the electric field between plates of capacitor is constant.

(ii) MEDIUM PARTLY AIR : $\qquad C = \dfrac{\in_0 A}{d - \left(t - \frac{t}{\in_r}\right)}$

When a di-electric slab of thickness t & relative permittivity $\in_r$ is introduced between the plates of an air capacitor, then the distance between the plates is effectively reduced by $\left(t - \dfrac{t}{\in_r}\right)$ irrespective of the position of the di-electric slab .

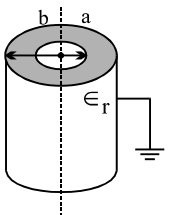


(iii) COMPOSITE MEDIUM : $\qquad C = \dfrac{\in_0 A}{\frac{t_1}{\in_{r1}} + \frac{t_2}{\in_{r2}} + \frac{t_3}{\in_{r3}}}$



4. CYLINDRICAL CAPACITOR :

It consist of two co-axial cylinders of radii a & b, the outer conductor is earthed. The di-electric constant of the medium filled in the space between the cylinder is $\in_r$. The capacitance per unit length is $C = \dfrac{2\pi \in_0 \in_r}{\ell n\left(\frac{b}{a}\right)} \dfrac{\text{Farad}}{\text{m}}$.



5. CONCEPT OF VARIATION OF PARAMETERS :

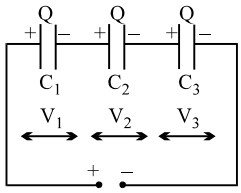
As capacitance of a parallel plate capacitor is $C = \dfrac{\in_0 kA}{d}$, if either of k, A or d varies in the region between the plates, we choose a small dc in between the plates and for total capacitance of system.

If all dC's are in series $\frac{1}{C_T} = \int \frac{dx}{\epsilon_0\, k(x)\, A(x)}$, If all dC's are in parallel $C_T = \int dC$

6. COMBINATION OF CAPACITORS :

(i) CAPACITORS IN SERIES :

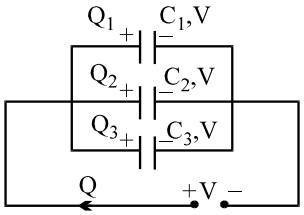
In this arrangement all the capacitors when uncharged get the same charge Q but the potential difference across each will differ (if the capacitance are unequal).



$$\frac{1}{C_{eq.}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \ldots\ldots + \frac{1}{C_n} .$$

(ii) CAPACITORS IN PARALLEL :

When one plate of each capacitor is connected to the positive terminal of the battery & the other plate of each capacitor is connected to the negative terminals of the battery, then the capacitors are said to be in parallel connection.



The capacitors have the same potential difference, V but the charge on each one is different (if the capacitors are unequal).

$C_{eq.} = C_1 + C_2 + C_3 + \ldots\ldots + C_n$.

7. ENERGY STORED IN A CHARGED CAPACITOR :

Capacitance C, charge Q & potential difference V ; then energy stored is

$U = \frac{1}{2} CV^2 = \frac{1}{2} QV = \frac{1}{2} \frac{Q^2}{C}$. This energy is stored in the electrostatic field set up in the di-electric medium between the conducting plates of the capacitor.

8. HEAT PRODUCED IN SWITCHING IN CAPACITIVE CIRCUIT

Due to charge flow always some amount of heat is produced when a switch is closed in a circuit which can be obtained by energy conservation as –

Heat = Work done by battery – Energy absorbed by capacitor.

9. SHARING OF CHARGES :

When two charged conductors of capacitance C_1 & C_2 at potential V_1 & V_2 respectively are connected by a conducting wire, the charge flows from higher potential conductor to lower potential conductor, until the potential of the two condensers becomes equal. The common potential (V) after sharing of charges;

$$V = \frac{\text{net charge}}{\text{net capacitance}} = \frac{q_1 + q_2}{C_1 + C_2} = \frac{C_1V_1 + C_2V_2}{C_1 + C_2} .$$

Charges after sharing $q_1 = C_1V$ & $q_2 = C_2V$. In this process energy is lost in the connecting wire as heat .

This loss of energy is $U_{initial} - U_{real} = \frac{C_1C_2}{2(C_1 + C_2)} (V_1 - V_2)^2$.

10. REMEMBER :

(i) The energy of a charged conductor resides outside the conductor in its EF, where as in a condenser it is stored within the condenser in its EF.

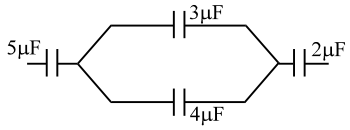
(ii) The energy of an uncharged condenser = 0 .

(iii) The capacitance of a capacitor depends only on its size & geometry & the di-electric between the conducting surface. (i.e. independent of the conductor, like, whether it is copper, silver, gold etc)

EXERCISE-I

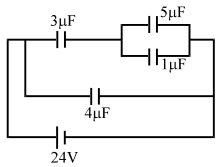
ONLY ONE OPTION IS CORRECT

1. If charge on left plane of the 5μF capacitor in the circuit segment shown in the figure is –20μC, the charge on the right plate of 3μF capacitor is



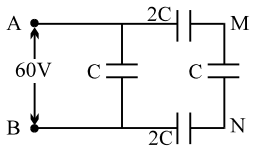
(A) +8.57 μC (B) –8.57 μC (C) +11.42 μC (D) –11.42 μC

2. In the circuit shown, the energy stored in 1μF capacitor is



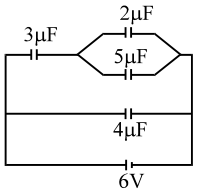
(A) 40 μJ (B) 64 μJ (C) 32 μJ (D) none

3. In the circuit shown, a potential difference of 60V is applied across AB. The potential difference between the point M and N is



(A) 10 V (B) 15 V (C) 20 V (D) 30 V

4. In the circuit shown in figure, the ratio of charges on 5μF and 4μF capacitor is :



(A) $\frac{4}{5}$ (B) $\frac{3}{5}$ (C) $\frac{3}{8}$ (D) $\frac{1}{2}$

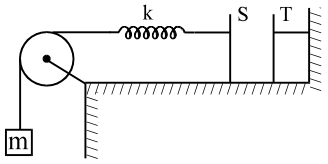
5. From a supply of identical capacitors rated 8 μF, 250 V, the minimum number of capacitors required to form a composite 16 μF, 1000 V is :

(A) 2 (B) 4 (C) 16 (D) 32

6. Two capacitor having capacitances 8 μF and 16 μF have breaking voltages 20 V and 80 V. They are combined in series. The maximum charge they can store individually in the combination is

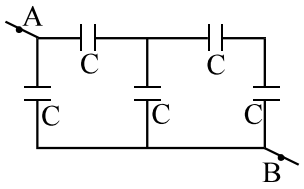
(A) 160 μC (B) 200 μC (C) 1280 μC (D) none of these

7. The plates S and T of an uncharged parallel plate capacitor are connected across a battery. The battery is then disconnected and the charged plates are now connected in a system as shown in the figure. The system shown is in equilibrium. All the strings are insulating and massless. The magnitude of charge on one of the capacitor plates is : [Area of plates = A]



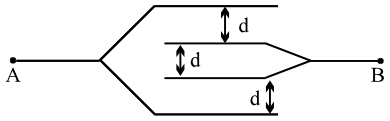
(A) $\sqrt{2mgA\in_0}$ (B) $\sqrt{\frac{4mgA\in_0}{k}}$ (C) $\sqrt{mgA\in_0}$ (D) $\sqrt{\frac{2mgA\in_0}{k}}$

8. What is the equivalent capacitance of the system of capacitors between A & B



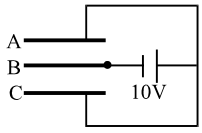
(A) $\frac{7}{6}$C (B) 1.6 C (C) C (D) None

9. Four metallic plates arearranged as shown in the figure. If the distance between each plate then capacitance of the given system between points A and B is (Given d << A)



(A) $\frac{\varepsilon_0 A}{d}$ (B) $\frac{2\varepsilon_0 A}{d}$ (C) $\frac{3\varepsilon_0 A}{d}$ (D) $\frac{4\varepsilon_0 A}{d}$

10. Three plates A, B and C each of area 0.1 m^2 are separated by 0.885 mm from each other as shown in the figure. A 10 V battery is used to charge the system. The energy stored in the system is

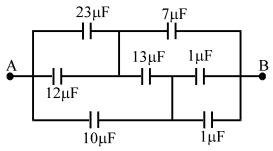


(A) 1 μJ (B) 10^{-1} μJ (C) 10^{-2} μJ (D) 10^{-3} μJ

11. A capacitor of capacitance C is initially charged to a potential difference of V volt. Now it is connected to a battery of 2V Volt with opposite polarity. The ratio of heat generated to the final energy stored in the capacitor will be

(A) 1.75 (B) 2.25 (C) 2.5 (D) $\frac{1}{2}$

12. Find the equivalent capacitance across A & B



(A) $\frac{28}{3}$ μf (B) $\frac{15}{2}$ μF (C) 15 μF (D) none

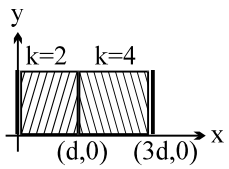
13. A parallel plate capacitor has an electric field of 10^5V/m between the plates. If the charge on the capacitor plate is 1μC, then the force on each capacitor plate is

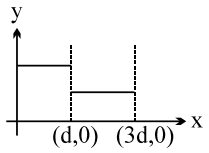
(A) 0.1 Nt (B) 0.05 Nt (C) 0.02 Nt (D) 0.01 Nt

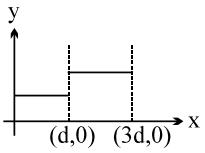
14. A capacitor is connected to a battery. The force of attraction between the plates when the separation between them is halved

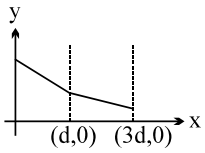
(A) remains the same (B) becomes eight times (C) becomes four times (D) becomes two times

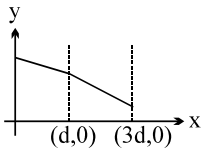
15. A parallel plate capacitor has two layers of dielectric as shown in figure. This capacitor is connected across a battery. The graph which shows the variation of electric field (E) and distance (x) from left plate.



(A) 

(B) 

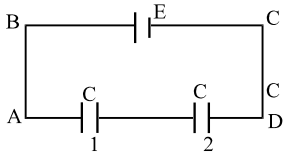
(C) 

(D) 

16. A capacitor stores 60 μC charge when connected across a battery. When the gap between the plates is filled with a dielectric, a charge of 120μC flows through the battery. The dielectric constant of the material inserted is :

(A) 1 (B) 2 (C) 3 (D) none

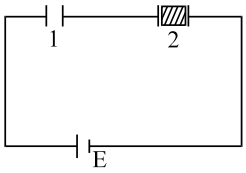
17. In the adjoining figure, capacitor (1) and (2) have a capacitance 'C' each. When the dielectric of dielectric consatnt K is inserted between the plates of one of the capacitor, the total charge flowing through battery is



(A) $\frac{KCE}{K+1}$ from B to C (B) $\frac{KCE}{K+1}$ from C to B

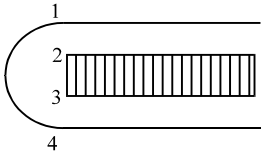
(C) $\frac{(K-1)CE}{2(K+1)}$ from B to C (D) $\frac{(K-1)CE}{2(K+1)}$ from C to B

18. Two identical capacitors 1 and 2 are connected in series to a battery as shown in figure. Capacitor 2 contains a dielectric slab of dielectric constant k as shown. Q_1 and Q_2 are the charges stored in the capacitors. Now the dielectric slab is removed and the corresponding charges are Q'_1 and Q'_2. Then



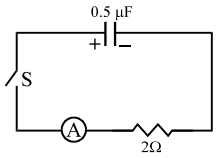
(A) $\frac{Q'_1}{Q_1} = \frac{k+1}{k}$ (B) $\frac{Q'_2}{Q_2} = \frac{k+1}{2}$ (C) $\frac{Q'_2}{Q_2} = \frac{k+1}{2k}$ (D) $\frac{Q'_1}{Q_1} = \frac{k}{2}$

19. Four identical plates 1, 2, 3 and 4 are placed parallel to each other at equal distance as shown in the figure. Plates 1 and 4 are joined together and the space between 2 and 3 is filled with a dielectric of dielectric constant k = 2. The capacitance of the system between 1 and 3 & 2 and 4 are C_1 and C_2 respectively. The ratio C_1/C_2 is :



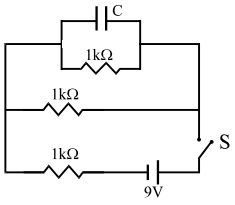
(A) $\frac{5}{3}$ (B) 1 (C) $\frac{3}{5}$ (D) $\frac{5}{7}$

20. A charged capacitor is allowed to discharge through a resistance 2Ω by closing the switch S at the instant t = 0. At time t = *l*n 2 μs, the reading of the ammeter falls half of its initial value. The resistance of the ammeter equal to



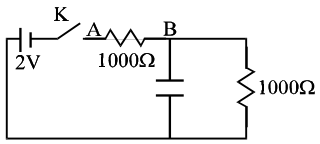
(A) 0 (B) 2Ω (C) ∞ (D) 2MΩ

21. A capacitor C = 100 μF is connected to three resistor each of resistance 1 kΩ and a battery of emf 9V. The switch S has been closed for long time so as to charge the capacitor. When switch S is opened, the capacitor discharges with time constant



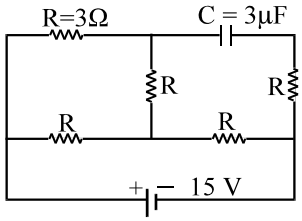
(A) 33 ms (B) 5 ms (C) 3.3 ms (D) 50 ms

22. In the circuit shown, when the key k is pressed at time t = 0, which of the following statements about current I in the resistor AB is true



(A) I = 2mA at all t (B) I oscillates between 1 mA and 2mA
(C) I = 1 mA at all t (D) At t = 0, I = 2mA and with time it goes to 1 mA

23. In the circuit shown, the cell is ideal, with emf = 15 V. Each resistance is of 3Ω. The potential difference across the capacitor is

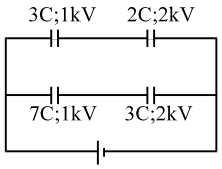


(A) zero (B) 9 V (C) 12 V (D) 15 V

24. Two capacitors C_1 and C_2 are connected in series, assume that $C_1 < C_2$. The equivalent capacitance of this arrangement is C, where
(A) $C < C_1/2$ (B) $C_1/2 < C < C_1$ (C) $C_1 < C < C_2$ (D) $C_2 < C < 2C_2$

25. Two capacitors C_1 and C_2 are connected in parallel, assume that $C_1 < C_2$. The equivalent capacitance of this arrangement is C, where
(A) $C < C_1/2$ (B) $C_1/2 < C < C_1$ (C) $C_1 < C < C_2$ (D) $C_2 < C < 2C_2$

26. The diagram shows four capacitors with capacitances and break down voltages as mentioned. What should be the maximum value of the external emf source such that no capacitor breaks down?



(A) 2.5 kV (B) 10 / 3 kV (C) 3 kV (D) 1 kV

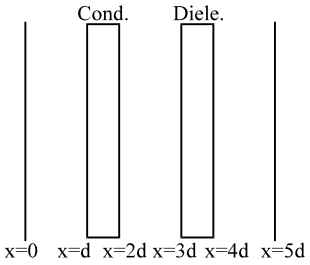
27. Three long concentric conducting cylindrical shells have radii R, 2R and $2\sqrt{2}$ R. Inner and outer shells are connected to each other. The capacitance across middle and inner shells per unit length is:

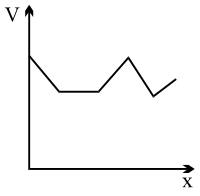
(A) $\frac{\frac{1}{3}\epsilon_0}{\ell n 2}$ (B) $\frac{6\pi\epsilon_0}{\ell n 2}$ (C) $\frac{\pi\epsilon_0}{2\ell n 2}$ (D) None

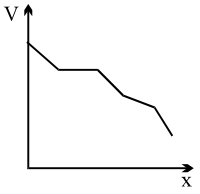
28. A capacitor of capacitance C is charged to a potential difference V from a cell and then disconnected from it. A charge +Q is now given to its positive plate. The potential difference across the capacitor is now

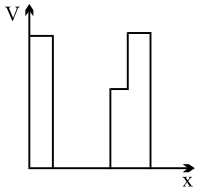
(A) V (B) $V + \frac{Q}{C}$ (C) $V + \frac{Q}{2C}$ (D) $V - \frac{Q}{C}$, if V < CV

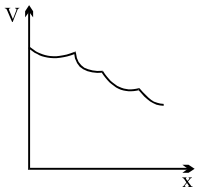
29. The distance between plates of a parallel plate capacitor is 5d. Let the positively charged plate is at x = 0 and negatively charged plate is at x = 5d. Two slabs one of conductor and other of a dielectric of equal thickness d are inserted between the plates as shown in figure. Potential versus distance graph will look like :



(A) 

(B) 

(C) 

(D) 

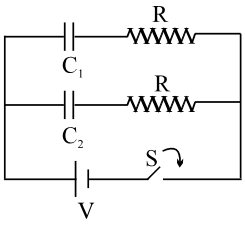
30. The distance between the plates of a charged parallel plate capacitor is 5 cm and electric field inside the plates is 200 Vcm^{-1}. An uncharged metal bar of width 2 cm is fully immersed into the capacitor. The length of the metal bar is same as that of plate of capacitor. The voltage across capacitor after the immersion of the bar is

(A) zero (B) 400 V (C) 600 V (D) 100 V

31. Condenser A has a capacity of 15 μF when it is filled with a medium of dielectric constant 15. Another condenser B has a capacity 1 μF with air between the plates. Both are charged separately by a battery of 100 V. After charging, both are connected in parallel without the battery and the dielectric material being removed. The common potential now is

(A) 400 V (B) 800 V (C) 1200 V (D) 1600 V

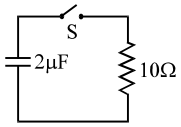
32. In the circuit shown in figure $C_1 = 2C_2$. Switch S is closed at time t = 0. Let i_1 and i_2 be the currents flowing through C_1 and C_2 at any time t, then the ratio i_1 / i_2



(A) is constant (B) increases with increase in time t

(C) decreases with increase in time t (D) first increases then decreases

33. In the R–C circuit shown in the figure the total energy of 3.6×10^{-3} J is dissipated in the 10 Ω resistor when the switch S is closed. The initial charge on the capacitor is

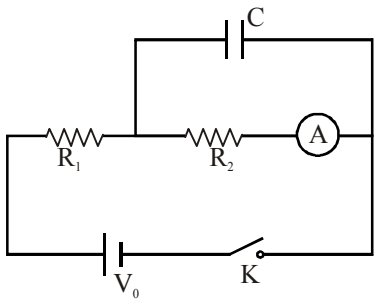


(A) 60 μC (B) 120 μC (C) $60\sqrt{2}$ μC (D) $\frac{60}{\sqrt{2}}$ μC

34. **Statement-1 :** The electrostatic force between the plates of a charged isolated capacitor decreases when dielectric fills whole space between plates.
Statement-2 : The electric field between the plates of a charged isolated capacitance decreases when dielectric fills whole space between plates.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

Paragraph for question nos. 35 to 37

In the connection shown in the figure initially the switch K is open and the capacitor is uncharged. Then the switch is closed and the capacitor is charged up to the steady state and the switch is opened again. Determine the values indicated by the ammeter. [Given : $V_0 = 30$ V, $R_1 = 10\ k\Omega$, $R_2 = 5k\Omega$]



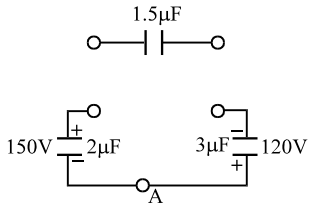
35. Just after closing the switch :
(A) 2 mA (B) 3 mA (C) 0 mA (D) None of these

36. A long time after the switch was closed :
(A) 2 mA (B) 3 mA (C) 6 mA (D) None of these

37. Just after reopening the switch :
(A) 2 mA (B) 3 mA (C) 6 mA (D) None of these

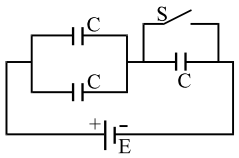
ONE OR MORE THAN ONE OPTION MAY BE CORRECT

38. Two capacitors of 2 μF and 3 μF are charged to 150 volt and 120 volt respectively. The plates of capacitor are connected as shown in the figure. A discharged capacitor of capacity 1.5 μF falls to the free ends of the wire. Then



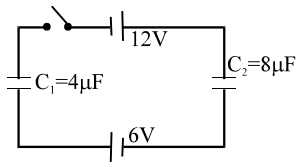
(A) charge on the 1.5 μF capacitors is 180 μC
(B) charge on the 2 μF capacitor is 120 μC
(C) positive charge flows through A from right to left.
(D) positive charge flows through A from left to right.

39. In the circuit shown, each capacitor has a capacitance C. The emf of the cell is E. If the switch S is closed



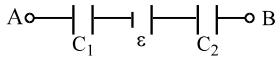
(A) positive charge will flow out of the positive terminal of the cell
(B) positive charge will enter the positive terminal of the cell
(C) the amount of charge flowing through the cell will be CE.
(D) the amount of charge flowing through the cell will be 4/3 CE.

40. In the circuit shown initially C_1, C_2 are uncharged. After closing the switch



(A) The charge on C_2 is greater that on C_1
(B) The charge on C_1 and C_2 are the same
(C) The potential drops across C_1 and C_2 are the same
(D) The potential drops across C_2 is greater than that across C_1

41. A circuit shown in the figure consists of a battery of emf 10 V and two capacitance C_1 and C_2 of capacitances 1.0 μF and 2.0 μF respectively. The potential difference $V_A - V_B$ is 5V

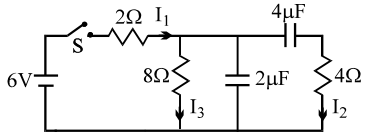


(A) charge on capacitor C_1 is equal to charge on capacitor C_2
(B) Voltage across capacitor C_1 is 5V.
(C) Voltage across capacitor C_2 is 10 V
(D) Energy stored in capacitor C_1 is two times the energy stored in capacitor C_2.

42. A capacitor C is charged to a potential difference V and battery is disconnected. Now if the capacitor plates are brought close slowly by some distance :
(A) some +ve work is done by external agent
(B) energy of capacitor will decrease
(C) energy of capacitor will increase
(D) none of the above

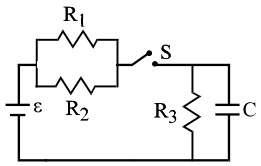
43. A parallel plate capacitor has a parallel slab of copper inserted between and parallel to the two plates, without touching the plates. The capacity of the capacitor after the introduction of the copper sheet is :
(A) minimum when the copper slab touches one of the plates.
(B) maximum when the copper slab touches one of the plates.
(C) invariant for all positions of the slab between the plates.
(D) greater than that before introducing the slab.

44. In the circuit shown in the figure, the switch S is initially open and the capacitor is initially uncharged. I_1, I_2 and I_3 represent the current in the resistance 2Ω, 4Ω and 8Ω respectively.



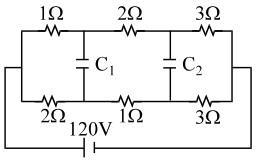
(A) Just after the switch S is closed, I_1 = 3A, I_2 = 3A and I_3 = 0
(B) Just after the switch S is closed, I_1 = 3A, I_2 = 0 and I_3 = 0
(C) long time after the switch S is closed, I_1 = 0.6 A, I_2 = 0 and I_3 = 0
(D) long after the switch S is closed, $I_1 = I_2 = I_3$ = 0.6 A.

45. The circuit shown in the figure consists of a battery of emf ε = 10 V ; a capacitor of capacitance C = 1.0 μF and three resistor of values R_1 = 2Ω, R_2 = 2Ω and R_3 = 1Ω. Initially the capacitor is completely uncharged and the switch S is open. The switch S is closed at t = 0.



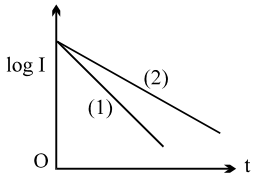
(A) The current through resistor R_3 at the moment the switch closed is zero.
(B) The current through resistor R_3 a long time after the switch closed is 5A.
(C) The ratio of current through R_1 and R_2 is always constant.
(D) The maximum charge on the capacitor during the operation is 5μC.

46. In the circuit shown in figure $C_1 = C_2$ = 2μF. Then charge stored in



(A) capacitor C_1 is zero
(B) capacitor C_2 is zero
(C) both capacitor is zero
(D) capacitor C_1 is 40 μC

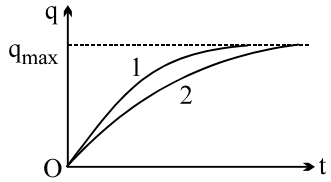
47. A capacitor of capacity C is charged to a steady potential difference V and connected in series with an open key and a pure resistor 'R'. At time t = 0, the key is closed. If I = current at time t, a plot of log I against 't' is as shown in (1) in the graph. Later one of the parameters i.e. V, R or C is changed keeping the other two constant, and the graph (2) is recorded. Then



(A) C is reduced
(B) C is increased
(C) R is reduced
(D) R is increased

Question No. 48 to 49 (2 questions)

The charge across the capacitor in two different RC circuits 1 and 2 are plotted as shown in figure.



48. Choose the correct statement(s) related to the two circuits.

(A) Both the capacitors are charged to the same charge.

(B) The emf's of cells in both the circuit are equal.

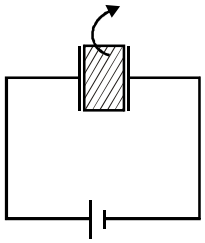
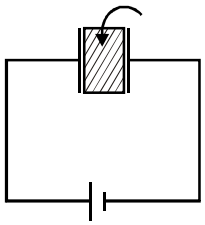
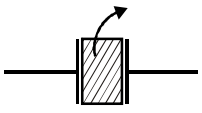
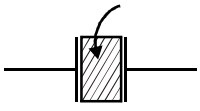
(C) The emf's of the cells may be different.

(D) The emf E_1 is more than E_2

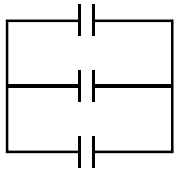
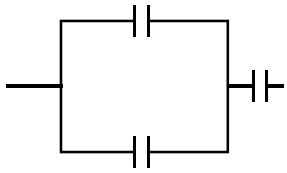
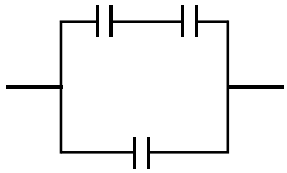
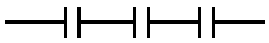
49. Identify the correct statement(s) related to the R_1, R_2, C_1 and C_2 of the two RC circuits.

(A) $R_1 > R_2$ if $E_1 = E_2$ (B) $C_1 < C_2$ if $E_1 = E_2$ (C) $R_1C_1 > R_2C_2$ (D) $\frac{R_1}{R_2} < \frac{C_2}{C_1}$

50.

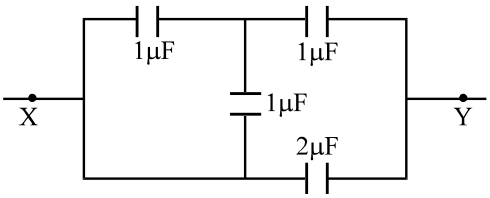
	Column-I		**Column-II**
(A)	 Removal of dielectric when battery is present	(P)	Potential difference between plates increases.
(B)	 Insertion of dielectric when battery is present	(Q)	Capacitance increases
(C)	 Removal of dielectric when battery is not present	(R)	Stored energy increases
(D)	 Insertion of dielectric when battery is not present	(S)	Charge present on plates decreases

51.

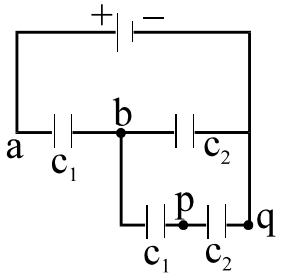
	Column-I		**Column-II** **Treat all capacitors as identical and initially uncharged**
(A)	If breakdown voltage of each capacitor is same which combination can withstand largest external voltage	(P)	
(B)	If same charge is supplied which combination acquires largest potential difference	(Q)	
(C)	If same potential difference is applied which combination can store maximum charge	(R)	
		(S)	

EXERCISE-II

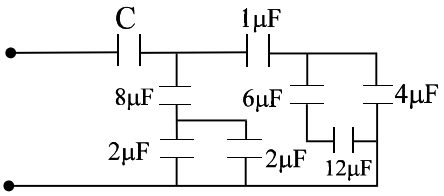
1. The figure shows a circuit consisting of four capacitors. Find the effective capacitance between X and Y.



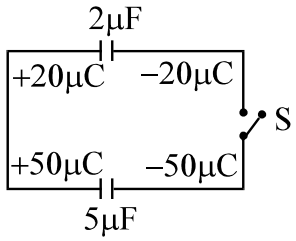
2. In the given network if potential difference between p and q is 2V and $C_2 = 3C_1$. Then find the potential difference between a & b.



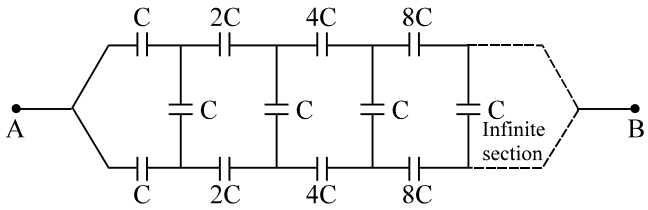
3. In the following circuit, the resultant capacitance between A and B is 1 μF. Find the value of C.



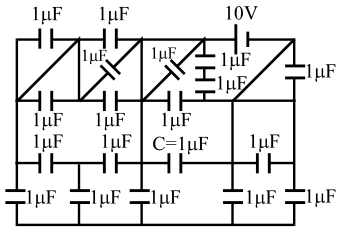
4. Find heat produced in the circuit shown in figure on closing the switch S.



5. Find the equivalent capacitance of the circuit between point A and B.

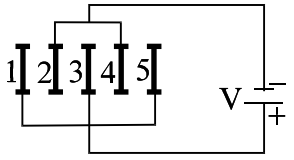


6. Find the charge on the capacitor C = 1 μF in the circuit shown in the figure.

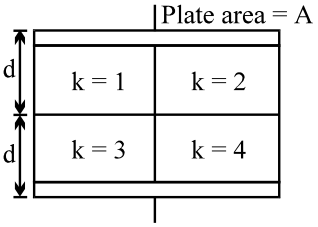


7. The plates of a parallel plate capacitor are given charges +4Q and –2Q. The capacitor is then connected across an uncharged capacitor of same capacitance as first one (= C). Find the final potential difference between the plates of the first capacitor.

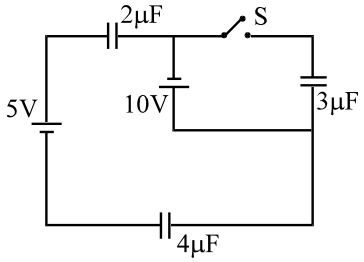
8. Five identical capacitor plates, each of area A, are arranged such that adjacent plates are at a distance 'd' apart, the plates are connected to a source of emf V as shown in figure. The charge on plate 1 is_____ and that on plate 4 is _____.



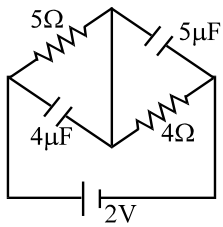
9. Find the capacitance of the system shown in figure.



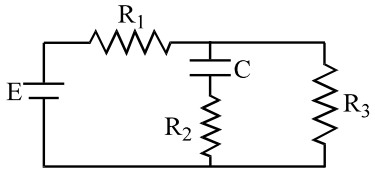
10. In the circuit shown in figure, find the amount of heat generated when switch s is closed.



11. Find the ratio between the energy stored in 5 μF capacitor to the 4 μF capacitor in the given circuit in steady state.



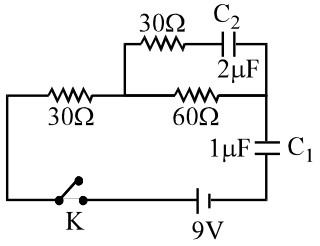
12. In the circuit shown here, at the steady state, the charge on the capacitor is ____.



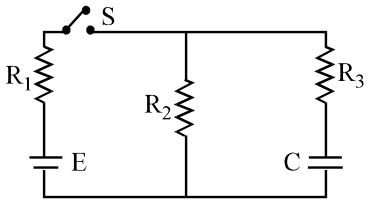
13. For the arrangement shown in the figure, the key is closed at t = 0. C_2 is initially uncharged while C_1 has a charge of 2μC.

(a) Find the current coming out of the battery just after switch is closed.

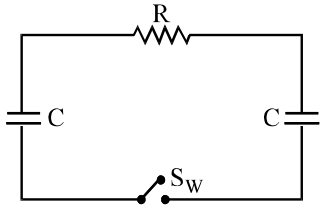
(b) Find the charge on the capacitors in the steady state condition.



14. In the circuit shown in figure $R_1 = R_2 = 6R_3 = 300\ M\Omega$, C = 0.01 μF and E = 10V. The switch is closed at t = 0, find (a) Charge on capacitor as a function of time. (b) energy of the capacitor at t = 20s.



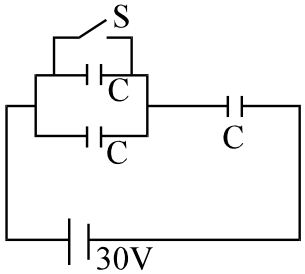
15. In the circuit shown in figure the capacitance of each capacitor is equal to C and resistance R. One of the capacitors was charge to a voltage V_0 and then at the moment t = 0 was shorted by means of the switch S. Find : (a) the current in the circuit as a function of time t. (b) the amount of generated heat.



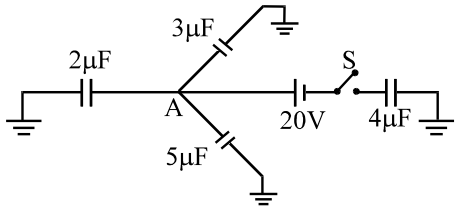
16. The capacitor each having capacitance C = 2μF are connected with a battery of emf 30 V as shown in figure. When the switch S is closed. Find

(a) the amount of charge flown through the battery (b) the heat generated in the circuit

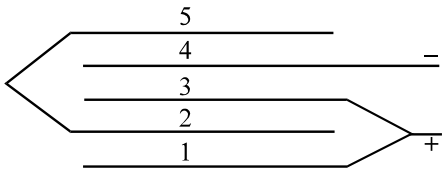
(c) the energy supplied by the battery (d) the amount of charge flown through the switch S



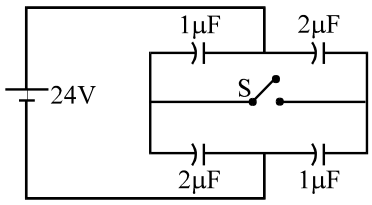
17. Three capacitors of 2μF, 3μF and 5μF are independently charged with batteries of emf's 5V, 20V and 10V respectively. After disconnecting from the voltage sources. These capacitors are connected as shown in figure with their positive polarity plates are connected to A and negative polarity is earthed. Now a battery of 20V and an uncharged capacitor of 4μF capacitance are connected to the junction A as shown with a switch S. When switch is closed, find : (a) the potential of the junction A. (b) final charges on all four capacitors.



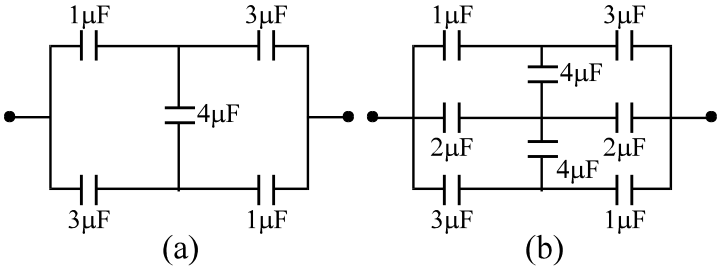
18. Five identical conducting plates 1, 2, 3, 4 & 5 are fixed parallel to and equdistant from each other (see figure). Plates 2 & 5 are connected by a conductor while 1 & 3 are joined by another conductor. The junction of 1 & 3 and the plate 4 are connected to a source of constant e.m.f. V_0. Find ;
(i) the effective capacity of the system between the terminals of the source.
(ii) the charges on plates 3 & 5.
Given d = distance between any 2 successive plates & A = area of either face of each plate.



19. The connections shown in figure are established with the switch S open. How much charge will flow through the switch if it is closed?



20. Find the equivalent capacitances of the combinations shown in the figure between the indicated points.



21. An insolated conductor initially free from charge is charged by repeated contacts with a plate which after each contact has a charge Q due to some mechanism. If q is the charge on the conductor after the first operation, prove that the maximum charge which can be given to the conductor in this way is $\frac{Qq}{Q-q}$.

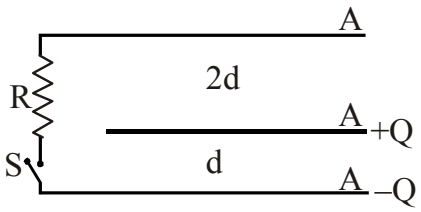
22. A potential difference of 300 V is applied between the plates of a plane capacitor spaced 1 cm apart. A plane parallel glass plate with a thickness of 0.5 cm and a plane parallel paraffin plate with a thickness of 0.5 cm are placed in the space between the capacitor plates find :
(i) Intensity of electric field in each layer.
(ii) The drop of potential in each layer.
(iii) The surface charge density of the charge on the capacitor plates. Given that : $k_{glass} = 6$, $k_{paraffin} = 2$

23. A parallel plate capacitor has plates with area A & separation d . A battery charges the plates to a potential difference of V_0. The battery is then disconnected & a di-electric slab of constant K & thickness d is introduced. Calculate the positive work done by the system (capacitor + slab) on the man who introduces the slab.

24. Two square metallic plates of 1 m side are kept 0.01 m apart, like a parallel plate capacitor, in air in such a way that one of their edges is perpendicular, to an oil surface in a tank filled with an insulating oil. The plates are connected to a battery of e.m.f. 500 volt . The plates are then lowered vertically into the oil at a speed of 0.001 m/s. Calculate the current drawn from the battery during the process.
[di-electric constant of oil = 11, $\in_0 = 8.85 \times 10^{-12}\ C^2/N^2m^2$]

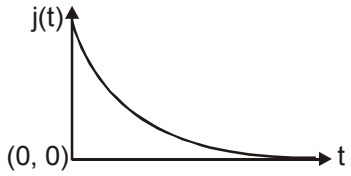
25. Three identical large metal plates of area A are small at distances d and 2d from each other. Top metal plate is uncharged, while other metal plates have charges +Q and –Q. Top and bottom metal plates are connected by switch S through a resistor of unknown resistance. What energy (in mJ) is dissipated in the resistor when switch is closed?

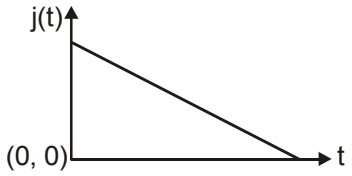
(Given : $\frac{\in_0 A}{d} = 6\ \mu F$, $Q = 60\ \mu C$)

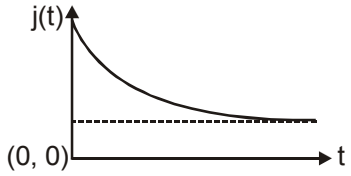


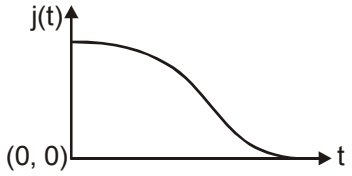
EXERCISE-III

1. An infinite line charge of uniform electric charge density λ lies along the axis of an electrically conducting infinite cylindrical shell of radius R. At time t = 0, the space inside the cylinder is filled with a material of permittivity ε and electrical conductivity σ. The electrical conduction in the material follows Ohm's law. Which one of the following graphs best describes the subsequent variation of the magnitude of current density j(t) at any point in the material? **[JEE Advance-2016]**

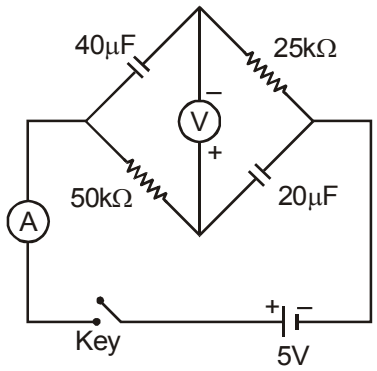
(A) 

(B) 

(C) 

(D) 

2. In the circuit shown below, the key is pressed at time t = 0. Which of the following statement(s) is(are) true?



(A) The voltmeter displays –5 V as soon as the key is pressed, and displays +5 V after a long time.

(B) The voltmeter will display 0 V at time t = ln 2 seconds. **[JEE Advance-2016]**

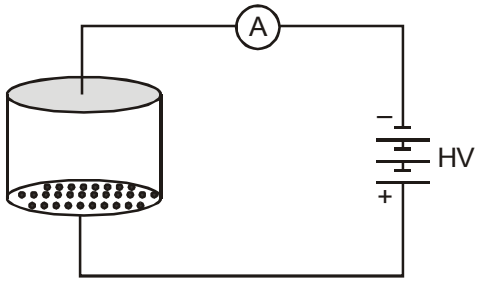
(C) The current in the ammeter becomes 1/e of the initial value after 1 second.

(D) The current in the ammeter becomes zero after a long time.

Paragraph (Q.3 to Q.4)

Consider an evacuated cylindrical chamber of height h having rigid conducting plates at the ends and an insulating curved surface as shown in the figure. A number of spherical balls made of a light weight and soft material and coated with a conducting material are placed on the bottom plate. The balls have a radius r << h. Now a high voltage source (HV) is connected across the conducting plates such that the bottom, plate is at $+V_0$ and the top plate at $-V_0$. Due to their conducting surface, the balls will get charged, will become equipotential with the plate and are repelled by it. The balls will eventaully collide with the top plate, where the coefficient of restitution can be taken to be zero due to the soft nature of the material of the balls. The electric field in the chamber can be considered to be that of a parallel plate capacitor. Assume that there are no collisions between the balls and the interaction between them is negligible. (Ignore gravity)

[JEE Advance-2016]



3. Which one of the following statements is correct?

(A) The balls will bounce back to the bottom plate carrying the opposite charge they went up with

(B) The balls will execute simple harmonic motion between the two plates

(C) The balls will bounce back to the bottom plate carrying the same charge they went up with

(D) The balls will stick to the top plate and remain there

4. The average current in the steady state registered by the ammeter in the circuit will be

(A) Proportional to $V_0^{1/2}$ (B) Proportional to V_0^2

(C) Proportional to the potential V_0 (D) Zero

Paragraph (Q.5 to Q.6)

Consider a simple RC circuit as shown in figure 1. **[JEE Advance-2017]**

Process 1 : In the circuit the switch S is closed at t = 0 and the capacitor is fully charged to voltage V_0 (i.e., charging continues for time T >> RC). In the process some dissipation (E_D) occurs across the resistance R. The amount of energy finally sotred in the fully charged capacitor is E_C.

Process 2 : In a different process the voltage is first set to $\frac{v_0}{3}$ and maintained for a charging time T >> RC.

Then the voltage is raised to $\frac{2v_0}{3}$ without discharging the capacitor and again maintained for a time T >> RC. The process is repeated one more time by raising the voltage to V_0 and the capacitor is charge to the some final voltage V_0 as in Process 1.

Thse two processes are depicted in Figure 2.

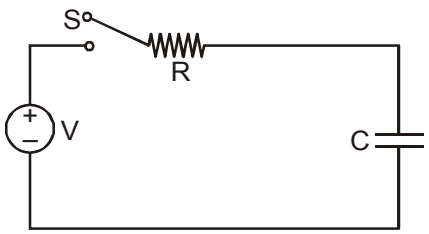


Figure 1

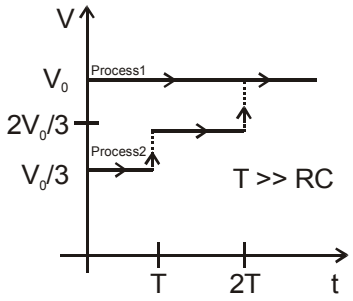


Figure 2

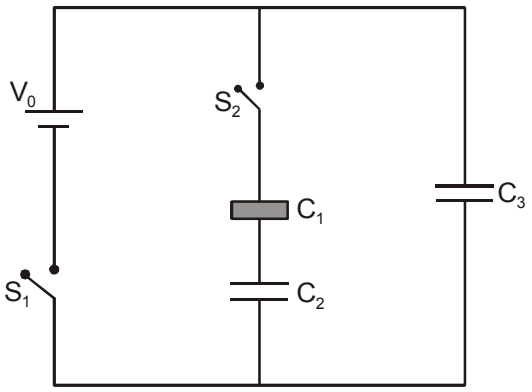
5. In Process 1, the energy stored in the capacitor E_C & heat dissipated across resistance E_D are related by

(A) $E_C = E_D$ (B) $E_C = 2E_D$ (C) $E_C = \frac{1}{2}E_D$ (D) $E_C = E_D \ln 2$

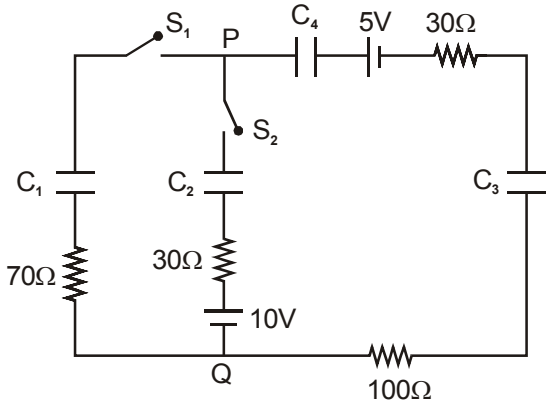
6. In Process 2, total energy dissipated across the resistance E_D is :

(A) $E_D = \frac{1}{3}\left(\frac{1}{2}CV_0^2\right)$ (B) $E_D = 3\left(\frac{1}{2}CV_0^2\right)$ (C) $E_D = \frac{1}{2}CV_0^2$ (D) $E_D = 3CV_0^2$

7. Three identical capacitors C_1, C_2 and C_3 have a capacitance of 1.0 μF each and they are uncharged initially. They are connected in a circuit as shown in the figure and C_1 is then filled completely with a dielectric material of relative permittivity $\in_r$. The cell electromotive force (emf) $V_0 = 8$ V. First the switch S_1 is closed while the switch S_2 is kept open. When the capacitor C_3 is fully charged, S_1 is opened and S_2 is closed simultaneously. When all the capacitors reach equilibrium, the charge on C_3 is found to be 5 μC. The value of $\in_r$ =_____. **[JEE-Advanced-2018]**



8. In the circuit shown, initially there is no charge on capacitors and keys S_1 and S_2 are open. The value of the capacitors are C_1 = 10 μF, C_2 = 30 μF and $C_3 = C_4$ = 80 μF. **[JEE-Advanced-2019]**



Which of the statement(s) is/are correct ?

(A) If key S_1 is kept closed for long time such that capacitors are fully charged, the voltage across the capacitor C_1 will be 4V.

(B) The key S_1 is kept closed for long time such that capacitors are fully charged. Now key S_2 is closed, at this time, the instantaneous current across 30Ω resistor (between points P and Q) will be 0.2 A (round off to 1st decimal place)

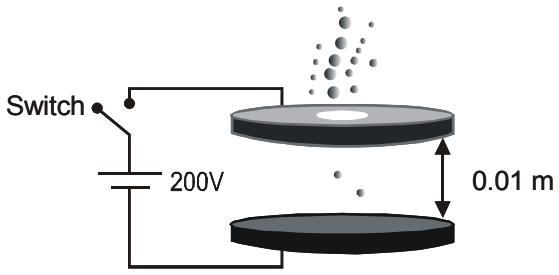
(C) If key S_1 is kept closed for long time such that capacitors are fully charged, the voltage difference between points P and Q wil be 10 V.

(D) At time t = 0, the key S_1 is closed, the instantaneous current in the closed circuit will be 25 mA.

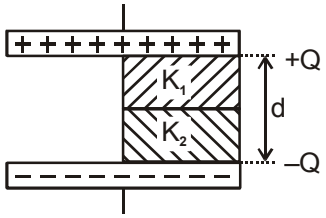
9. A parallel plate capacitor of capacitance C has spacing d between two plates having area A. The region between the plates is filled with N dielectric layers. parallel to its plates, each with thickness $\delta = \frac{d}{N}$. The dielectric constant of the mth layer is $K_m = K\left(1+\frac{m}{N}\right)$. For a very large N (> 10^3), the capacitance C is $\alpha\left(\frac{K\in_0 A}{d \ln 2}\right)$. The value of α will be.... [$\in_0$ is the permittivity of free space] **[JEE-Advanced-2019]**

10. Two large circular discs separated by a distance of 0.01 m are connected to a battery via a switch as shown in the figure. Charged oil drops of density 900 kg m^{-3} are released through a tiny hole at the center of the top disc. Once some oil drops achieve terminal velocity, the switch is closed to apply a voltage of 200 V across the discs. As a result, an oil drop of radius 8 × 10^{-7} m stops moving vertically and floats between the discs. The number of electrons present in this oil drop is _____.

(neglect the buoyancy force, take acceleration due to gravity = 10 ms^{-2} and charge on an electron (e) = 1.6 × 10^{-19} C) **[JEE-Advanced-2020]**



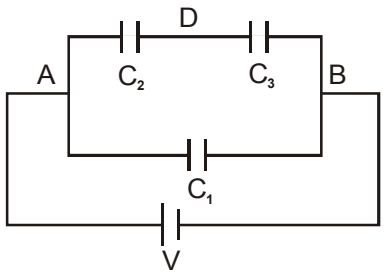
11. A parallel-plate capacitor with plate area A has separation d between the plates. Two dielectric slabs of dielectric constant K_1 and K_2 of same area A/2 and thickness d/2 are inserted in the space between the plates. The capacitance will be given by : **[JEE-Mains Online-2021]**



(A) $\frac{\varepsilon_0 A}{d}\left(\frac{1}{2}+\frac{K_1+K_2}{K_1K_2}\right)$ (B) $\frac{\varepsilon_0 A}{d}\left(\frac{1}{2}+\frac{K_1K_2}{2(K_1+K_2)}\right)$

(C) $\frac{\varepsilon_0 A}{d}\left(\frac{1}{2}+\frac{2(K_1+K_2)}{K_1K_2}\right)$ (D) $\frac{\varepsilon_0 A}{d}\left(\frac{1}{2}+\frac{K_1K_2}{K_1+K_2}\right)$

12. Three capacitors C_1 = 2 μF, C_2 = 6 μF & C_3 = 12 μF are connected as shown in figure. Find the ratio of the charges on capacitors C_1, C_2 & C_3 respectively : **[JEE-Mains Online-2021]**



(A) 2 : 3 : 3 (B) 3 : 4 : 4 (C) 1 : 2 : 2 (D) 2 : 1 : 1

13. The material filled between the plates of a parallel plate capacitor has resistivity 200 Ωm. The value of capacitance of the capacitor is 2 pF. If a potential difference of 40 V is applied across the plates of the capacitor, then the value of leakage current flowing out of the capacitor is : (given the value of relative permitivity of material is 50) **[JEE-Mains Online-2021]**

(A) 9.0 mA (B) 9.0 μA (C) 0.9 mA (D) 0.9 μA

14. A capacitor is connected to a 20 V battery through a resistance of 10 Ω. It is found that the potential difference across the capacitor rises to 2 V in 1 μs. The capacitance of the capacitor is_____ μF.

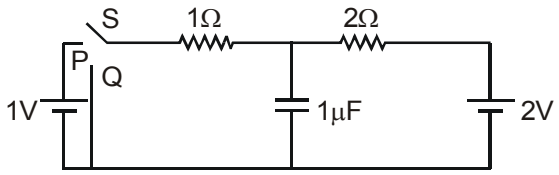
Given $\ln\left(\frac{10}{9}\right) = 0.105$ **[JEE-Mains Online-2021]**

(A) 1.85 (B) 9.52 (C) 0.95 (D) 0.105

15. A parallel plate capacitor of capacitance 200 μF is connected to a battery of 200 V. A dielectric slab of dielectric constant 2 is now inserted into the space between plates of capacitor while the battery remain conected. The change in the electrostatic energy in the capacitor will be_____J. **[JEE-Mains Online-2021]**

Question Stem for Question Nos. 16 and 17

In the circuit shown below, the switch S is connected to position P for a long time so that the charge on the capacitor becomes q_1 μC. Then S is switched to position Q. After a long time, the charge on the capacitor is q_2 μC. **[JEE-Advanced-2021]**



16. The magnitude of q_1 is _____.

17. The magnitude of q_2 is _____.

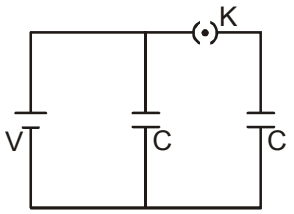
18. Two identical thin metal plates has charge q_1 and q_2 respectively such that $q_1 > q_2$. The plates were brought close to each other to form a parallel plate capacitor of capacitance C. The potential difference between them is : **[JEE-Mains Online-2022]**

(A) $\frac{(q_1 + q_2)}{C}$ (B) $\frac{(q_1 - q_2)}{C}$ (C) $\frac{(q_1 - q_2)}{2C}$ (D) $\frac{2(q_1 - q_2)}{C}$

19. Two capacitors, each having capacitance 40 μF are connected in series. The space between one of the capacitors is filled with dielectric material of dielectric constant K such that the equivalence capacitance of the system became 24 μF. The value of K will be : **[JEE-Mains Online-2022]**

(A) 1.5 (B) 2.5 (C) 1.2 (D) 3

20. A source of potential difference V is connected to the combination of two identical capacitors as shown in the figure. When key 'K' is closed, the total energy stored across the combination is E_1. Now key 'K' is opened and dielectric of dielectric constant 5 is introduced between the plates of th capacitors. The total energy stored across the combination is now E_2. The ratio E_1 / E_2 will be **[JEE-Mains Online-2022]**



(A) $\frac{1}{10}$ (B) $\frac{2}{5}$ (C) $\frac{5}{13}$ (D) $\frac{5}{26}$

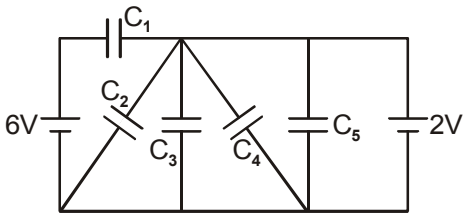
21. The total charge on the system of capacitance C_1 = 1 μF, C_2 = 2 μF, C_3 = 4 μF and C_4 = 3 μF connected in parallel is (Assume a battery of 20V is connected to the combination) **[JEE-Mains Online-2022]**

(A) 200 μC (B) 200 C (C) 10 μC (D) 10 C

22. A slab of dielectric constant K has the same crosssectional area as the plates of a parallel plate capacitor and thickness $\frac{3}{4}d$, where d is the separation of the plates. The capacitance of the capacitor when the slab is inserted between the plates will be :
(Given C_o = capacitance of capacitor with air as medium between plates.) **[JEE-Mains Online-2022]**

(A) $\frac{4KC_0}{3+K}$ (B) $\frac{3KC_0}{3+K}$ (C) $\frac{3+K}{4KC_0}$ (D) $\frac{K}{4+K}$

23. In the following circuit C_1 = 12 μF, $C_2 = C_3$ = 4 μF and $C_4 = C_5$ = 2 μF. The charge stored in C_3 is_____ μC. **[JEE-Advanced-2022]**



24. A medium having dielectric constant K > 1 fills the space between the plates of a parallel plate capacitor. The plates have large area, and the distance between them is d. The capacitor is connected to a battery of voltage V, as shown in Figure (a). Now, both the plates are moved by a distance of $\frac{d}{2}$ from their original positions, as shown in Figure (b). In the process of going from the configuration depicted in Figure (a) to that in Figure (b), which of the following statement(s) is(are) correct? **[JEE-Advanced-2022]**

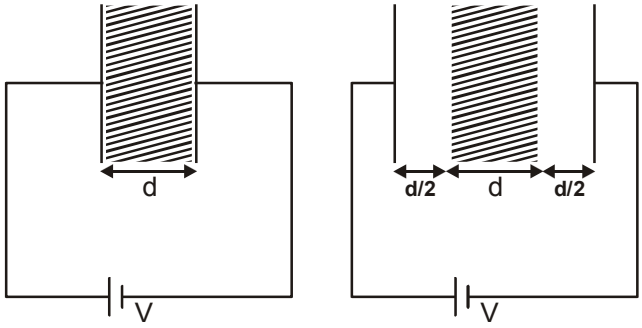


Figure (a) Figure (b)

(A) The electric field inside the dielectric material is reduced by a factor of 2K.

(B) The capacitance is decreased by a factor of $\frac{1}{K+1}$.

(C) The voltage between the capacitor plates is increased by a factor of (K + 1).

(D) The work done in the process **DOES NOT** depend on the presence of the dielectric material.

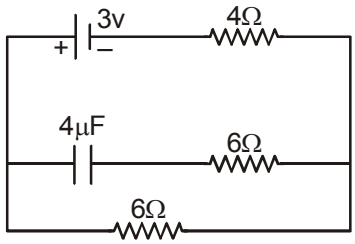
25. The distance between two plates of a capacitor is d and its capacitance is C_1 , when air is the medium between the plates. If a metal sheet of thickness $\frac{2d}{3}$ and of the same area as plate is introduced between the plates, the capacitance of the capacitor becomes C_2. The ratio $\frac{C_2}{C_1}$ is : **[JEE-Mains Online-2023]**

(A) 4 : 1 (B) 2 : 1 (C) 3 : 1 (D) 1 : 1

26. A capacitor of capacitance C is charged to a potential V. The flux of the electric field through a closed surface enclosing the positive plate of the capacitor is : **[JEE-Mains Online-2023]**

(A) $\frac{CV}{2\varepsilon_0}$ (B) $\frac{2CV}{\varepsilon_0}$ (C) Zero (D) $\frac{CV}{\varepsilon_0}$

27. In the network shown below, the charge accumulated in the capacitor in steady state will be **[JEE-Mains Online-2023]**

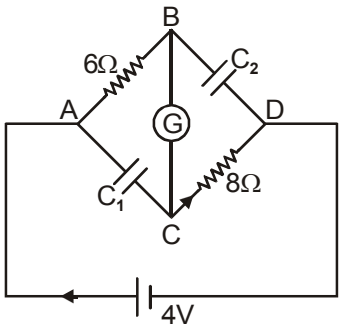


(A) 7.2 μC (B) 10.3 μC (C) 1.8 μC (D) 12 μC

28. A parallel plate capacitor of capacitance 2F is charged to a potential V, the energy stored in the capacitor is E_1. The capacitor is now connected to another uncharged identical capacitor in parallel combination. The energy stored in the combination is E_2. The ratio E_2 / E_1 is : **[JEE-Mains Online-2023]**

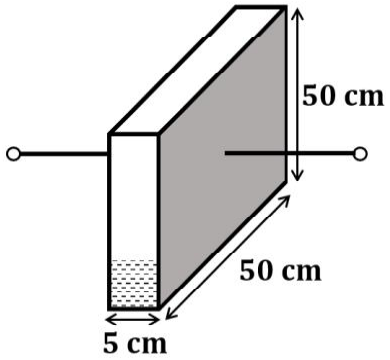
(A) 1 : 2 (B) 2 : 1 (C) 1 : 4 (D) 2 : 3

29. In this figure the resistance of the coil of galvanometer G is 2Ω. The emf of the cell is 4 V. The ratio of potential difference across C_1 and C_2 is : **[JEE-Mains Online-2023]**



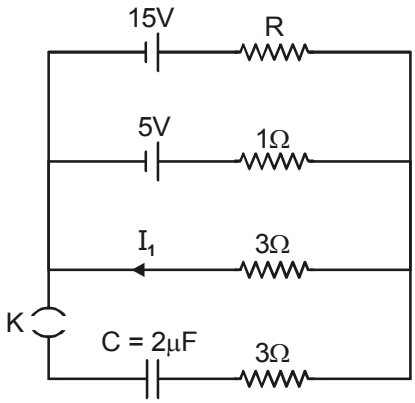
(A) 1 (B) $\frac{3}{4}$ (C) $\frac{5}{4}$ (D) $\frac{4}{5}$

30. A container has a base of 50 cm × 5 cm and height 50 cm, as shown in the figure. It has two parallel electrically conducting walls each of area 50 cm × 50 cm. The remaining walls of the container are thin and non-conducting. The container is being filled with a liquid of dielectric constant 3 at a uniform rate of 250 $cm^3 s^{-1}$. What is the value of the capacitance of the container after 10 seconds?
[Given : Permittivity of free space $\varepsilon_0 = 9 \times 10^{-12} C^2N^{-1}m^{-2}$, the effects of the non-conducting walls on the capacitance are negligible] **[JEE-Advanced-2023]**



(A) 27 pF (B) 63 pF (C) 81 pF (D) 135 pF

31. In a circuit shown in the figure, the capacitor C is initially uncharged and the key K is open. In this condition, a current of 1 A flows through the 1 Ω resistor. The key is closed at time $t = t_0$. Which of the following statement(s) is(are) correct? [Given: $e^{-1} = 0.36$] **[JEE-Advanced-2023]**

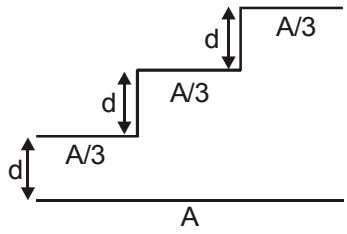


(A) The value of the resistance R is 3Ω.
(B) For $t < t_0$, the value of current I_1 is 2A.
(C) At $t = t_0 + 7.2$ μs, the current in the capacitor is 0.6A.
(D) For $t \to \infty$, the charge on the capacitor is 12μC.

32. A bulb and a capacitor are connected in series across an ac supply. A dielectric is then placed between the plates of the capacitor. The glow of the bulb. **[JEE-Mains Online-2024]**
(A) increases (B) remains same (C) decreases (D) becomes zero

33. A capacitor is made of a flat plate of area A and a second plate having a stair-like structure as shown in figure. If the area of each stair is $\frac{A}{3}$ and the height is d. the capacitance of the arrangement is :

[JEE-Mains Online-2024]



(A) $\frac{13 \in_0 A}{17 d}$ (B) $\frac{18 \in_0 A}{11 d}$ (C) $\frac{11 \in_0 A}{20 d}$ (D) $\frac{11 \in_0 A}{18 d}$

34. A capacitor has air as dielectric medium and two conducting plates of area 12 cm^2 and they are 0.6 cm apart. When a alab of dielectric having area 12 cm^2 and 0.6 cm thickness is inserted between ithe plates, one of the conducting plates has to be moved by 0.2 cm to keep the capacitance same as in previous case. The dielectric constant of the slab is : (Given $\varepsilon_0 = 8.834 \times 10^{-12}$ F/m) **[JEE-Mains Online-2024]**

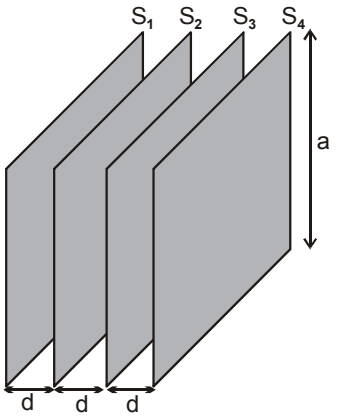
(A) 1.33 (B) 1.50 (C) 0.66 (D) 1

35. A parallel plate capacitor of capacitance 12.5 pF is charged by a battery connected between its plates to potential difference of 12.0 V. The battery is now disconnected and a dielectric slab ($\in_r = 6$) is inserted between the plates. The change in its potential energy after inserting the dielectric slab is_____ $\times 10^{-12}$ J.

[JEE-Mains Online-2024]

36. Three capacitors of capacitances 25 μF, 30 μF and 45 μF are connected in parallel to a supply of 100 V. Energy stored in the above combination is E. When these capacitors are connected in series to the same supply, the stored energy is $\frac{9}{x}E$. The value of x is _____. **[JEE-Mains Online-2024]**

37. Four identical thin, square metal sheets, S_1, S_2, S_3 and S_4, each of side 'a' are kept parallel to each other with equal distance d(<< a) between them, as shown in the figure. Let $C_0 = \varepsilon_0 a^2/d$, where ε_0 is the permittivity of free space.



Match the quantities mentioned in List-I with their values in List-II and choose the correct option.

	List-I		**List-II**
(P)	The capacitance between S_1 and S_4, with S_2 and S_3 not connected, is	(A)	$3C_0$
(Q)	The capacitance between S_1 and S_4, with S_2 shorted to S_3, is	(B)	$\frac{C_0}{2}$
(R)	The capacitance between S_1 and S_3, with S_2 shorted to S_4, is	(C)	$\frac{C_0}{3}$
(S)	The capacitance between S_1 and S_2, with	(D)	$\frac{2C_0}{3}$
	S_3 shorted to S_1, and S_2 shorted to S_4, is	(5)	$2C_0$

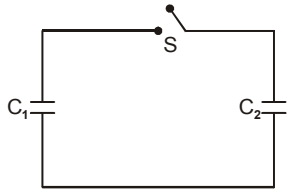
[JEE-Advanced-2024]

(A) P → 3 ; Q → 2 ; R → 4 ; S → 5 (B) P → 2 ; Q → 3 ; R → 2 ; S → 1
(C) P → 3 ; Q → 2 ; R → 4 ; S → 1 (D) P → 3 ; Q → 2 ; R → 2 ; S → 5

38. A parallel-plate capacitor of capacitance 40 μF is connected to a 100 V power supply. Now the intermediate space between the plates is filled with a dielectric material of dielectric constant K = 2. Due to the introduction of dielectric material, the extra charge and the change in the electrostatic energy in the capacitor, respectively, are **[JEE-Mains Online-2025]**

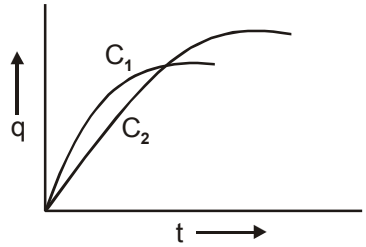
(A) 2 mC and 0.2 J (B) 8 mC and 2.0 J (C) 4 mC and 0.2 J (D) 2 mC and 0.4 J

39. A capacitor, C_1 = 6 μF is charged to a potential difference of V_0 = 5V using a 5V battery. The battery is removed and another capacitor, C_2 = 12 μF is inserted in place of the battery. When the switch 'S' is closed, the charge flows between the capacitors for some time until equilibrium condition is reached. What are the charges (q_1 and q_2) on the capacitors C_1 and C_2 when equilibrium condition is reached.



(A) q_1 = 15 μC, q_2 = 30 μC (B) q_1 = 30 μC, q_2 = 15 μC **[JEE-Mains Online-2025]**
(C) q_1 = 10 μC, q_2 = 20 μC (D) q_1 = 20 μC, q_2 = 10 μC

40. Two capacitors C_1 and C_2 are connected in parallel to a battery. Charge-time graph is shown below for the two capacitors. The energy stored with them are U_1 and U_2, respectively. Which of the given statements is true ? **[JEE-Mains Online-2025]**



(A) $C_1 > C_2, U_1 > U_2$ (B) $C_2 > C_1, U_2 < U_1$ (C) $C_1 > C_2, U_1 < U_2$ (D) $C_2 > C_1, U_2 > U_1$

41. Which one of the following is the correct dimensional formula for the capacitance in F ? M, L, T and C stand for unit of mass, length, time and charge, **[JEE-Mains Online-2025]**

(A) $[F] = [C^2M^{-2}L^2T^2]$ (B) $[F] = [CM^{-2}L^{-2}T^{-2}]$

(C) $[F] = [CM^{-1}L^{-2}T^2]$ (D) $[F] = [C^2M^{-1}L^{-2}T^2]$

42. Consider a parallel plate capacitor of area A (of each plate) and separation 'd' between the plates. If E is the electric field and ε_0 is the permittivity of free space between the plates, then potential energy stored in the capacitor is **[JEE-Mains Online-2025]**

(A) $\frac{1}{2}\varepsilon_0 E^2 Ad$ (B) $\frac{3}{4}\varepsilon_0 E^2 Ad$ (C) $\frac{1}{4}\varepsilon_0 E^2 Ad$ (D) $\varepsilon_0 E^2 Ad$

43. A parallel plate capacitor of capacitance 1 μF is charged to a potential difference of 20 V. The distance between plates is 1 μm. The energy density between plates of capacitor is : **[JEE-Mains Online-2025]**

(A) 1.8×10^3 J/m^3 (B) 2×10^{-4} J/m^3 (C) 2×10^2 J/m^3 (D) 1.8×10^5 J/m^3

44. A parallel plate capacitor was made with two rectangular plates, each with a length of ℓ = 3 cm and breath of b = 1 cm. The distance between the plates is 3 μm. Out of the following, which are the ways to increase the capacitance by a factor of 10 ? **[JEE-Mains Online-2025]**

(A) ℓ = 30 cm, b = 1 cm, d = 1 μm (B) ℓ = 3 cm, b = 1 cm, d = 30 μm

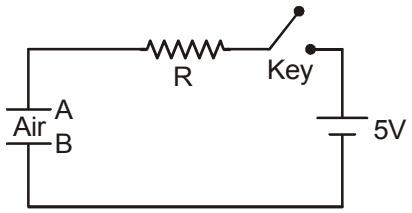
(C) ℓ = 6 cm, b = 5 cm, d = 3 μm (D) ℓ = 1 cm, b = 1 cm, d = 10 μm

(E) ℓ = 5 cm, b = 2 cm, d = 1 μm

Choose the correct answer from the options given below :

(A) C and E only (B) B and D only (C) A only (D) C only

45. Identify the valid statements relevant to the given circuit at the instant when the key is closed.



A. There will be no current through resistor R.

B. There will be maximum current in the connecting wires. **[JEE-Mains Online-2025]**

C. Potential difference between the capacitor plates A and B is minimum.

D. Charge on the capacitor plates is minimum.

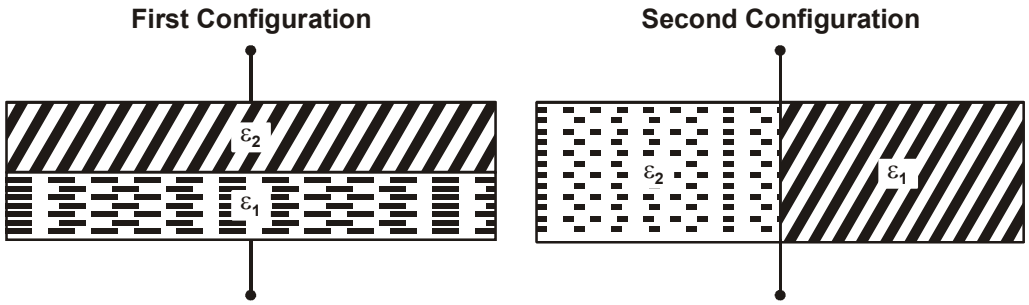
Choose the correct answer from the options given below :

(A) C, D only (B) B, C, D only (C) A, C only (D) A, B, D only

46. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : Net dipole moment of a polar linear isotropic dielectric substance is not zero even in the absence of an external electric field.
Reason (R) : In absence of an external electric field, the different permanent dipoles of a polar dielectric substance are oriented in random directions. In the light of the above statements, choose the most appropriate answer from the options given below **[JEE-Mains Online-2025]**
(A) **(A)** is correct but **(R)** is not correct
(B) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**
(C) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**
(D) **(A)** is not correct but **(R)** is correct

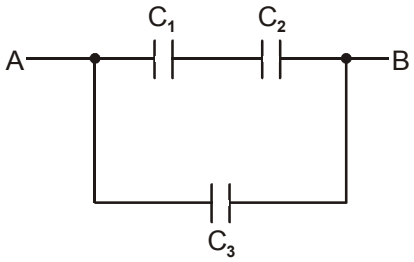
47. Using a battery, a 100 pF capacitor is charged to 60 V and then the battery is removed. After that, a second uncharged capacitor is connected to the first capacitor in parallel. If the final voltage across the second capacitor is 20 V, its capacitance is : (in pF) **[JEE-Mains Online-2025]**
(A) 600 (B) 200 (C) 400 (D) 10

48. A parallel plate capacitor is filled equally (half) with two dielectrics of dielectric constant ε_1 and ε_2, as shown in figures. The distance between the plates is d and area of each plate is A. If capacitance in first configuration and second configuration are C_1 and C_2 respectively, then $\frac{C_1}{C_2}$ is : **[JEE-Mains Online-2025]**



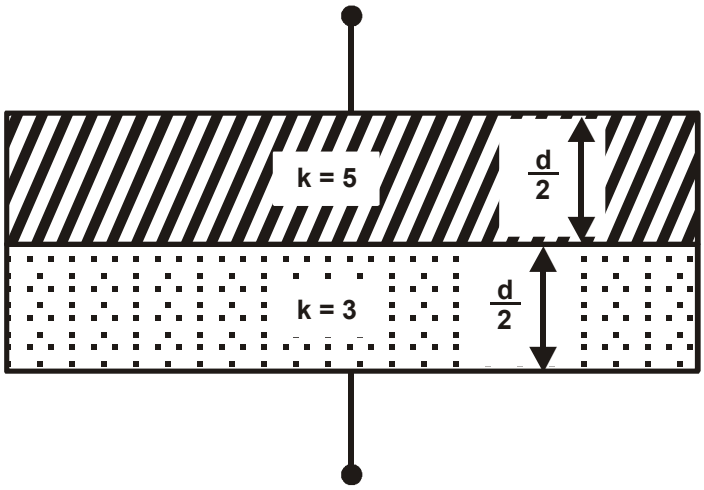
(A) $\frac{\varepsilon_1\varepsilon_2^2}{(\varepsilon_1+\varepsilon_2)^2}$ (B) $\frac{4\varepsilon_1\varepsilon_2}{(\varepsilon_1+\varepsilon_2)^2}$ (C) $\frac{\varepsilon_1\varepsilon_2}{\varepsilon_1+\varepsilon_2}$ (D) $\frac{\varepsilon_0(\varepsilon_1+\varepsilon_2)}{2}$

49. Three parallel plate capacitors C_1, C_2 and C_3 each of capacitance 5 μF are connected as shown in figure. The effective capacitance between points A and B, when the space between the parallel plates of C_1 capacitor is filled with a dielectric medium having dielectric constant of 4, is : **[JEE-Mains Online-2025]**

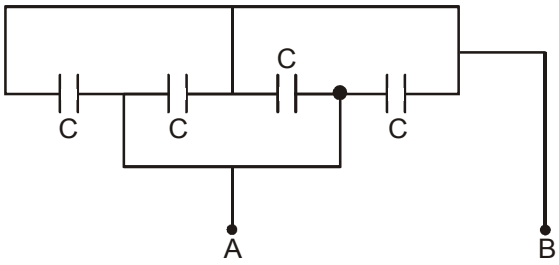


(A) 22.5 μF (B) 7.5 μF (C) 9 μF (D) 30 μF

50. Space between the plates of a parallel plate capacitor of plate area 4 cm^2 and separation of (d) 1.77 mm, is filled with uniform dielectric materials with dielectric constants (3 and 5) as shown in figure. Another capacitor of capacitance 7.5 pF is connected in parallel with it. The effective capacitance of this combination is _____ pF. (Given $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m) **[JEE-Mains Online-2025]**

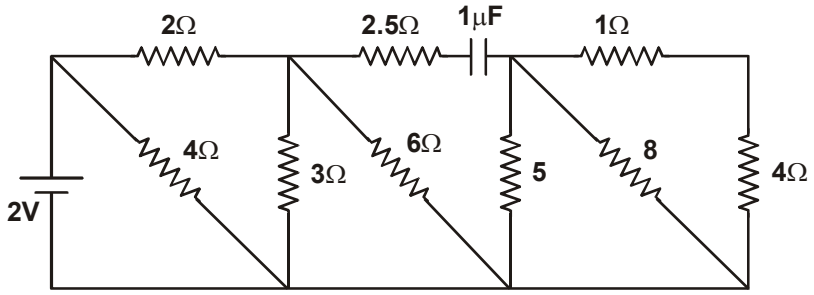


51. Four capacitor each of capacitance 16 μF are connected as shown in the figure. The capacitance between points A and B is : _____ (in μF) **[JEE-Mains Online-2025]**



52. A parallel plate capacitor has charge 5×10^{-6} C. A dielectric slab is inserted between the plates and almost fills the space between the plates. If the induced charge on one face of the slab is 4×10^{-6} C then the dielectric constant of the slab is _____. **[JEE-Mains Online-2025]**

53. The net current flowing in the given circuit is _____ A. **[JEE-Mains Online-2025]**

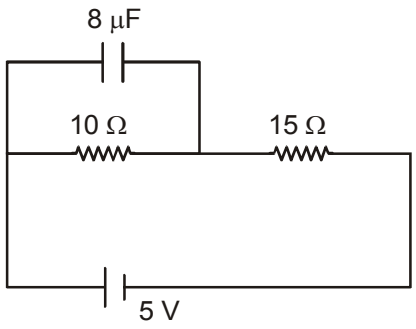


54. A parallel plate capacitor consisting of two circular plates of radius 10 cm is being charged by a constant current of 0.15 A. If the rate of change of potential difference between the plates is 7×10^8 V/s then the integer value of the distance between the parallel plates is **[JEE-Mains Online-2025]**

$\left(\text{Take, } \epsilon_0 = 9 \times 10^{-12} \frac{\text{F}}{\text{m}}, \pi = \frac{22}{7}\right)$ _____ μm.

55. At steady state the charge on the capacitor, as shown in the circuit below, is_____ μC.

[JEE-Mains Online-2025]



ANSWER KEY

EXERCISE-I

ONLY ONE OPTION IS CORRECT

1.	A	**2.**	C	**3.**	D	**4.**	C	**5.**	D	**6.**	A	**7.**	A
8.	B	**9.**	B	**10.**	B	**11.**	B	**12.**	B	**13.**	B	**14.**	C
15.	A	**16.**	C	**17.**	D	**18.**	C	**19.**	B	**20.**	A	**21.**	D
22.	D	**23.**	C	**24.**	B	**25.**	D	**26.**	A	**27.**	B	**28.**	C
29.	B	**30.**	C	**31.**	B	**32.**	B	**33.**	B	**34.**	D	**35.**	C
36.	A	**37.**	A										

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

38.	ABC	**39.**	AD	**40.**	B	**41.**	AD	**42.**	B	**43.**	CD	**44.**	B
45.	ABCD	**46.**	BD	**47.**	B	**48.**	AC	**49.**	D				

50. A – S ; B – Q, R ; C – P, R ; D – Q

51. A – S ; B – S ; C – P

EXERCISE-II

1. $\frac{8}{3}\mu F$ **2.** 30 V **3.** $\frac{32}{23}\mu F$ **4.** 0.00 **5.** C

6. $10\,\mu C$ **7.** $\frac{3Q}{2C}$ **8.** $\frac{A\in_0 V}{d}$, $-\frac{2A\in_0 V}{d}$

9. $\frac{25}{24}\frac{\varepsilon_0 A}{d}$ **10.** $150\,\mu J$ **11.** 0.8 **12.** $C\left(\frac{E}{R_1+R_3}\right)R_3$

13. (a) $\frac{7}{50}A$ or $\frac{11}{50}A$, (b) $Q_1 = 9\mu C$, $Q_2 = 0$ **14.** (a) $q = 0.05(1 - e^{-t/2})\,\mu C$; (b) $0.125\,\mu J$

15. (a) $I = \frac{V_0}{R}\,e^{-2t/Rc}$; (b) $\frac{1}{4}CV_0^2$ **16.** (a) $20\,\mu C$, (b) 0.3 mJ, (c) 0.6 mJ. (d) $60\,\mu C$

17. (a) $\frac{100}{7}$ volts; (b) $28.56\,\mu C$, $42.84\,\mu C$, $71.4\,\mu C$, $22.88\,\mu C$

18. (i) $\frac{5}{3}\left(\frac{\in_0 A}{d}\right)$; (ii) $Q_3 = \frac{4}{3}\left(\frac{\in_0 AV_a}{d}\right)$, $Q_5 = \frac{2}{3}\left(\frac{\in_0 AV_a}{d}\right)$

19. $12\mu C$ **20.** (a) $\frac{11}{6}\,\mu F$; (b) $\frac{11}{4}\,\mu F$ **21.** ☺

22. (i) 1.5×10^4 V/m, 4.5×10^4 V/m, (ii) 75 V, 225 V, (iii) 8×10^{-7} C/m^2

23. $W = \frac{1}{2}C_0V_0^2\left(1 - \frac{1}{K}\right)$ **24.** 4.425×10^{-9} A **25.** 0.10 J

EXERCISE-III

1.	A	**2.**	ABCD	**3.**	A	**4.**	B	**5.**	A
6.	A	**7.**	1.50	**8.**	ABD	**9.**	1.00 (0.99 to 1.01)		
10.	6.00	**11.**	D	**12.**	C	**13.**	C	**14.**	C
15.	4.00	**16.**	1.30 -1.40	**17.**	0.60 - 0.72	**18.**	C	**19.**	A
20.	C	**21.**	A	**22.**	A	**23.**	8.00	**24.**	B
25.	C	**26.**	A	**27.**	A	**28.**	A	**29.**	D
30.	B	**31.**	A,B,C,D	**32.**	A	**33.**	D	**34.**	B
35.	750	**36.**	86.00	**37.**	C	**38.**	C	**39.**	C
40.	D	**41.**	D	**42.**	A	**43.**	A	**44.**	A
45.	B	**46.**	D	**47.**	B	**48.**	B	**49.**	C
50.	15.00	**51.**	64.00	**52.**	5.00	**53.**	1.00	**54.**	1320
55.	16.00								